32. Word Rules

Year 4 · **Supplementary Worksheet** — extra practice & worked breakdowns

Worked Examples

Example 1 — A simple letter-shift:

In this code, each letter is replaced by the letter 1 place later in the alphabet (wrapping back round to A after Z), so CAT becomes DBU. Using the same rule, what does SWAN become?

Answer: S→T, W→X, A→B, N→O. Answer: **TXBO**.

Example 2 — Decoding a reversed word:

In this code, the word is written backwards (back to front), so CAT becomes TAC. Using the same rule, the word TAOG was made from an original word. What was the original word?

Answer: Reading TAOG backwards gives **GOAT**.

Example 3 — Rotating the first letter to the end:

In this code, the first letter of the word moves to the end, and every other letter shifts one place to the left, so CARS becomes ARSC. Using the same rule, what does DRAGON become?

Answer: The D moves from the front to the back: RAGON + D. Answer: **RAGOND**.

Example 4 — Watch out: letters can wrap round more than once:

In this code, each letter is replaced by the letter 3 places later in the alphabet (wrapping back round to A after Z), so CAT becomes FDW. Using the same rule, what does LYNX become? (Hint: two of these letters wrap all the way round past Z.)

Answer: L→O, Y→Z→A→B (wraps round), N→Q, X→Y→Z→A (wraps round). Answer: **OBQA**.

Example 5 — A combined rule, decoding:

In this code, each letter is first replaced by the letter 1 place later in the alphabet (wrapping back round to A after Z), and then the whole shifted word is written backwards, so CAT becomes UBD. Using the same rule, the word FUBSJQ was made from an original word. What was the original word?

Answer: First reverse FUBSJQ to undo the backwards step: QJSBUF. Then shift each letter 1 place EARLIER to undo the shift: Q→P, J→I, S→R, B→A, U→T, F→E. Answer: **PIRATE**.

Foundation

15 questions

1. In this code, each letter is replaced by the letter 1 place later in the alphabet (wrapping back round to A after Z), so CAT becomes DBU. Using the same rule, what does FROG become?

2. In this code, each letter is replaced by the letter 2 places later in the alphabet (wrapping back round to A after Z), so CAT becomes ECV. Using the same rule, what does CROW become?

3. In this code, the word is written backwards (back to front), so CAT becomes TAC. Using the same rule, the word TAOG was made from an original word. What was the original word?

4. In this code, the word is written backwards (back to front), so CAT becomes TAC. Using the same rule, what does SEAL become?

5. In this code, each letter is replaced by the letter 1 place later in the alphabet (wrapping back round to A after Z), so CAT becomes DBU. Using the same rule, what does HARE become?

6. In this code, each letter is replaced by the letter 2 places later in the alphabet (wrapping back round to A after Z), so CAT becomes ECV. Using the same rule, the word OQNG was made from an original word. What was the original word?

7. In this code, each letter is replaced by the letter 1 place later in the alphabet (wrapping back round to A after Z), so CAT becomes DBU. Using the same rule, what does TOAD become?

8. In this code, each letter is replaced by the letter 2 places later in the alphabet (wrapping back round to A after Z), so CAT becomes ECV. Using the same rule, what does WASP become? A) YCUR B) YCRR C) WCUR D) YCUC

9. In this code, the word is written backwards (back to front), so CAT becomes TAC. Using the same rule, what does VOLE become?

10. In this code, each letter is replaced by the letter 1 place later in the alphabet (wrapping back round to A after Z), so CAT becomes DBU. Using the same rule, the word LJXJ was made from an original word. What was the original word?

11. In this code, the word is written backwards (back to front), so CAT becomes TAC. Using the same rule, the word HTOM was made from an original word. What was the original word?

12. In this code, each letter is replaced by the letter 2 places later in the alphabet (wrapping back round to A after Z), so CAT becomes ECV. Using the same rule, what does BEAR become?

13. In this code, each letter is replaced by the letter 1 place later in the alphabet (wrapping back round to A after Z), so CAT becomes DBU. Using the same rule, what does DEER become? A) EFFS B) EFFF C) EYFS D) EFKS

14. In this code, each letter is replaced by the letter 2 places later in the alphabet (wrapping back round to A after Z), so CAT becomes ECV. Using the same rule, what does SWAN become?

15. In this code, the word is written backwards (back to front), so CAT becomes TAC. Using the same rule, select ALL of these words that become a DIFFERENT real English word once the code is applied. A) TRAP B) GOAT C) EWE D) STAR E) DRUM

Standard

20 questions

1. In this code, each letter is replaced by the letter 3 places later in the alphabet (wrapping back round to A after Z), so CAT becomes FDW. Using the same rule, what does BERRY become?

2. In this code, the first letter of the word moves to the end, and every other letter shifts one place to the left, so CARS becomes ARSC. Using the same rule, what does ROBIN become?

3. In this code, the first and last letters of the word swap places, and the letters in between stay the same, so CARS becomes SARC. Using the same rule, what does TIGER become?

4. In this code, each letter is replaced by the letter 3 places later in the alphabet (wrapping back round to A after Z), so CAT becomes FDW. Using the same rule, what does PANDA become?

5. In this code, the first letter of the word moves to the end, and every other letter shifts one place to the left, so CARS becomes ARSC. Using the same rule, what does CYGNET become?

6. In this code, the first and last letters of the word swap places, and the letters in between stay the same, so CARS becomes SARC. Using the same rule, what does WIZARD become?

7. In this code, each letter is replaced by the letter 3 places later in the alphabet (wrapping back round to A after Z), so CAT becomes FDW. Using the same rule, what does CASTLE become?

8. In this code, the first letter of the word moves to the end, and every other letter shifts one place to the left, so CARS becomes ARSC. Using the same rule, the word RAGOND was made from an original word. What was the original word?

9. In this code, the first and last letters of the word swap places, and the letters in between stay the same, so CARS becomes SARC. Using the same rule, the word EIRATP was made from an original word. What was the original word?

10. In this code, each letter is replaced by the letter 3 places later in the alphabet (wrapping back round to A after Z), so CAT becomes FDW. Using the same rule, the word EDVNHW was made from an original word. What was the original word?

11. In this code, the first letter of the word moves to the end, and every other letter shifts one place to the left, so CARS becomes ARSC. Using the same rule, what does JUNGLE become?

12. In this code, the first and last letters of the word swap places, and the letters in between stay the same, so CARS becomes SARC. Using the same rule, what does CANDLE become?

13. In this code, each letter is replaced by the letter 3 places later in the alphabet (wrapping back round to A after Z), so CAT becomes FDW. Using the same rule, what does BRIDGE become?

14. In this code, the first letter of the word moves to the end, and every other letter shifts one place to the left, so CARS becomes ARSC. Using the same rule, the word SLANDI was made from an original word. What was the original word?

15. In this code, each letter is replaced by the letter 3 places later in the alphabet (wrapping back round to A after Z), so CAT becomes FDW. Using the same rule, what does WINDOW become? A) ZLQGRZ B) LLQGRZ C) ZCQGRZ D) ZLQGRH

16. In this code, the first and last letters of the word swap places, and the letters in between stay the same, so CARS becomes SARC. Using the same rule, what does GARLIC become? A) CARLIG B) CAGLIG C) CARLPG D) CARLIT

17. In this code, the first letter of the word moves to the end, and every other letter shifts one place to the left, so CARS becomes ARSC. Using the same rule, what does GOBLIN become?

18. In this code, each letter is replaced by the letter 3 places later in the alphabet (wrapping back round to A after Z), so CAT becomes FDW. Using the same rule, what does VOYAGE become?

19. In this code, the first letter of the word moves to the end, and every other letter shifts one place to the left, so CARS becomes ARSC. Using the same rule, select ALL of these words that become a DIFFERENT real English word once the code is applied. A) STOP B) TARS C) SNAP D) CARE E) TEAM

20. In this code, the first and last letters of the word swap places, and the letters in between stay the same, so CARS becomes SARC. Using the same rule, what does PUDDLE become?

Hard

5 questions

1. In this code, each letter is replaced by the letter 4 places later in the alphabet (wrapping back round to A after Z), so CAT becomes GEX. Using the same rule, what does FALCON become?

2. In this code, the first letter of the word is written twice at the start, and then the rest of the word follows as normal, so CAT becomes CCAT. Using the same rule, what does PARROT become?

3. In this code, each letter is first replaced by the letter 1 place later in the alphabet (wrapping back round to A after Z), and then the whole shifted word is written backwards, so CAT becomes UBD. Using the same rule, what does SQUARE become? A) FSBVRT B) FSBQRT C) ASBVRT D) FGBVRT

4. In this code, each letter is first replaced by the letter 1 place later in the alphabet (wrapping back round to A after Z), and then the whole shifted word is written backwards, so CAT becomes UBD. Using the same rule, select ALL of these coded words that were made from a real English word. A) QPU B) QPX C) FDJ D) FDQ E) UBC

5. In this code, each letter is replaced by the letter 4 places later in the alphabet (wrapping back round to A after Z), so CAT becomes GEX. Using the same rule, the word GMVGPI was made from an original word. What was the original word?

© PassFrame Tuition — for registered student use